Page Tables

1 Core Idea

Definition

A **page table** maps virtual page numbers to physical page frame numbers.

The MMU uses the page table during address translation. The virtual address is split into:

- a **virtual page number**: high-order bits;
- an **offset**: low-order bits.

Page Table as a Function

$$\boxed{\text{page table}(\text{virtual page number}) = \text{page frame number}}$$

2 Address Split

With a 16-bit virtual address and a 4-KB page size:

Address part	Meaning
Upper 4 **bits**	Select one of 16 virtual pages.
Lower 12 **bits**	Select byte offset 0 to 4095 inside that page.

Why 12 Offset Bits?

4 KB $= 4096 = 2^{12}$, so 12 bits are needed to identify any byte inside one page.

Different splits are possible. More page-number bits means more pages. More offset bits means larger pages.

3 Translation Steps

Step	Action
1	Split the virtual address into page number and offset.
2	Use the virtual page number as an index into the page table.
3	Read the page table entry.
4	If the page is present, get the page frame number.
5	Combine page frame number with the unchanged offset.

Address Construction

$$\boxed{\text{physical address} = \text{page frame number} \mid\mid \text{offset}}$$

Here || means concatenation: the page frame number becomes the high-order part of the physical address, and the offset remains the low-order part.

4 Inside the MMU

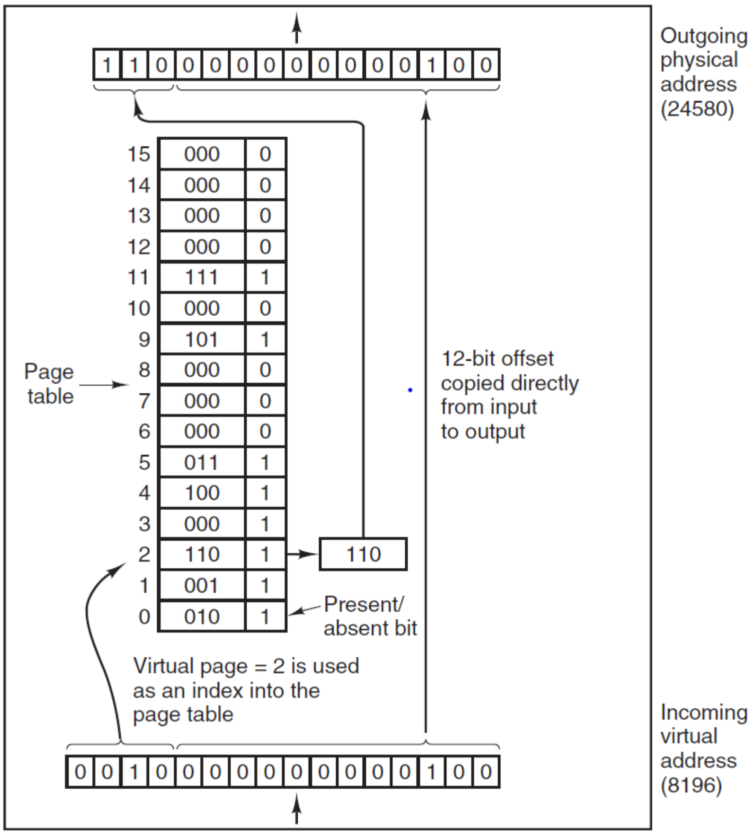

Figure 3-10. The internal operation of the MMU with 16 4-KB pages.

In the figure, the incoming virtual address is 8196. In binary it is:

$$0010\ 0000\ 0000\ 0100$$

The upper 4 bits are:

$$0010_2 = 2$$

So virtual page 2 is used as the page-table index. The page table entry for virtual page 2 contains frame number:

$$110_2 = 6$$

The lower 12 bits are the offset:

$$0000\ 0000\ 0100_2 = 4$$

Therefore:

$$\text{physical address} = 6 \times 4096 + 4 = 24580$$

Key Detail

The offset is copied directly from the virtual address to the physical address. Only the page number is replaced.

5 When the Page Is Absent

The page table entry also says whether the page is currently in RAM.

Present/absent bit	Meaning
1	Entry is valid; translation can continue.
0	Page is not in memory; page fault occurs.

Page Fault

If the present/absent bit is 0, the MMU traps to the operating system instead of producing a physical address.

6 Page Table Entry

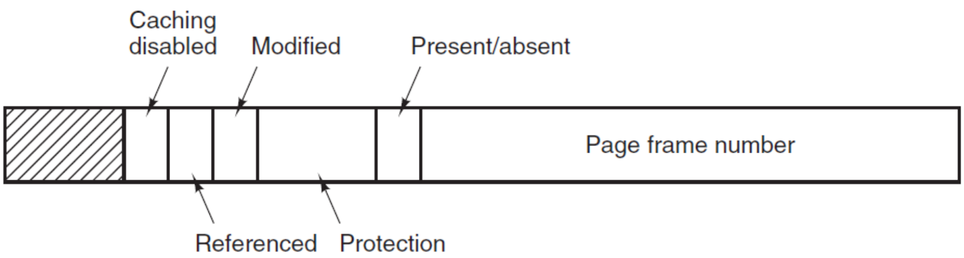

Figure 3-11. A typical page table entry.

Page Table Entry

A **page table entry (PTE)** stores the information needed to translate and control access to one virtual page.

The exact layout is machine dependent, but common fields appear on many systems.

7 Important Fields

Field	Purpose
Page frame number	Tells which physical frame holds the page. This is the main output of translation.
Present/absent bit	Says whether the page is currently in physical memory.
Protection bits	Control allowed access, such as read, write, and execute.
Modified bit	Set when the page is written; also called the dirty bit.
Referenced bit	Set when the page is read or written. Helps choose pages for replacement.
Caching disabled bit	Turns off caching for special pages, such as memory-mapped device registers.

8 Protection Bits

Protection bits define which operations are legal.

Protection style	Meaning
Simple form	One bit: read/write or read-only.
R/W/X form	Separate bits allow reading, writing, and executing.

Access Violation

If a process tries an operation not allowed by the protection bits, the hardware traps to the operating system.

9 Modified and Referenced Bits

Modified bit

The **Modified bit** is set when a page is written.

State	Replacement action
Modified/dirty	Must be written back to disk before eviction.
Clean	Can be discarded because the disk copy is already valid.

Referenced bit

The **Referenced bit** is set when a page is read or written. It helps the OS estimate whether a page is actively being used.

Replacement Use

When choosing a page to evict, the OS prefers pages that have not been referenced recently.

10 Caching Disabled Bit

Some virtual pages map to device registers instead of ordinary memory. In that case, caching can be dangerous.

Reason

If a device register is cached, the CPU may repeatedly read an old cached value instead of the current device value.

The caching disabled bit forces the hardware to fetch the value from the device or memory location every time.

11 What Is Not in the Page Table

The disk address of an absent page is usually not stored in the hardware page table.

Information	Stored where?
Hardware translation data	Page table entry.
Disk location of absent page	Operating-system software tables.

Reason

The page table stores what the hardware needs for address translation. Page-fault recovery information is managed by the operating system.

12 Summary

Concept	Meaning
Page table	Maps virtual page numbers to physical frame numbers.
Virtual page number	Index into the page table.
Offset	Copied unchanged into the physical address.
Present/absent bit	Determines whether translation succeeds or causes a page fault.
Modified bit	Tells whether the page must be written back before eviction.
Referenced bit	Helps page replacement algorithms identify active pages.
Protection bits	Control read, write, and execute permissions.